

**CAFFEINE & CADAVER**

MBBS BIOCHEMISTRY NOTES SERIES

PYQ 2010–2025

MBBS BIOCHEMISTRY · MODULE 02

Carbohydrate Metabolism

Glycolysis · TCA Cycle · HMP Shunt · Glycogen · Gluconeogenesis

SUBJECT

Biochemistry — Paper 1

EXAM RELEVANCE

Group A · B · C · D

PYQ FREQUENCY

12+ times (2010–2025)

★ MBBS PYQ APPEARANCES – YEARS

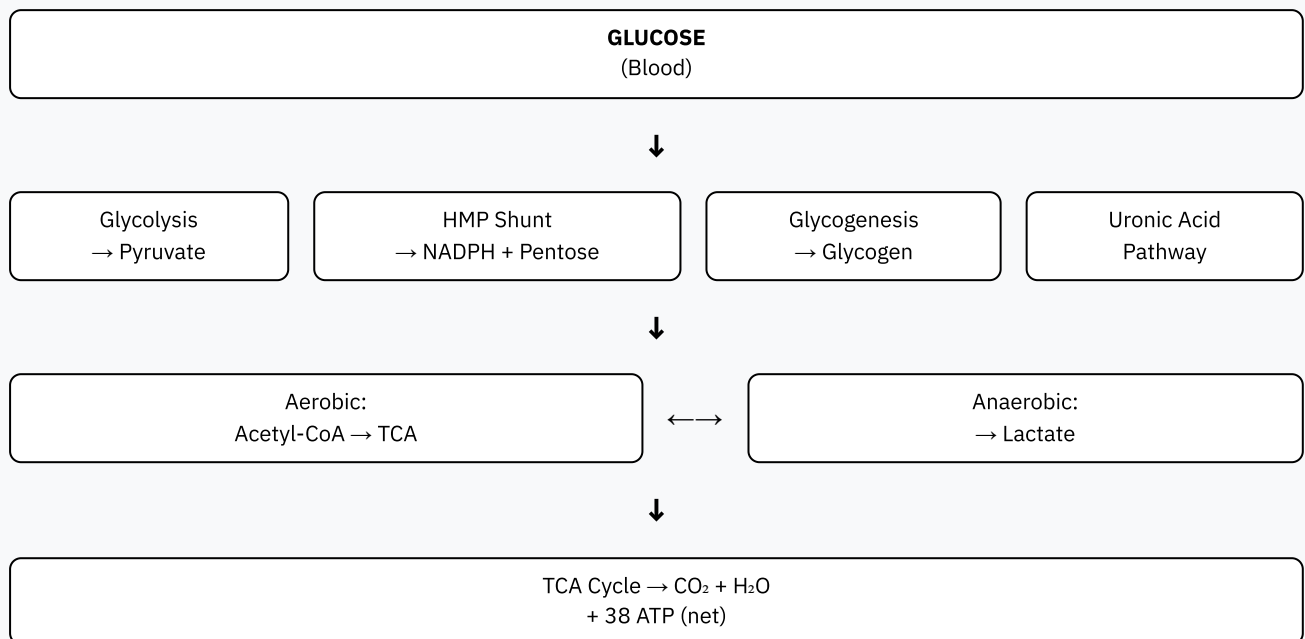
2010 2010-S 2012-S 2013 2014-S 2015-S 2016 2016-S 2017 2017-S 2018-S 2019
2019-S 2021 2022 2023 2025

01 The Big Picture

🔗 WHY THIS MATTERS CLINICALLY

Carbohydrate metabolism is the **central hub of energy production** in the body. Every pathway here connects to a clinical disease — Diabetes (glycolysis regulation), G6PD deficiency (HMP shunt), Von Gierke's disease (glycogen storage), Lactic acidosis (pyruvate dehydrogenase deficiency). Your examiner will test this as Group A almost every year because it demands flowcharts, enzyme knowledge, and clinical correlation all in one answer.

// MASTER OVERVIEW – FATE OF GLUCOSE

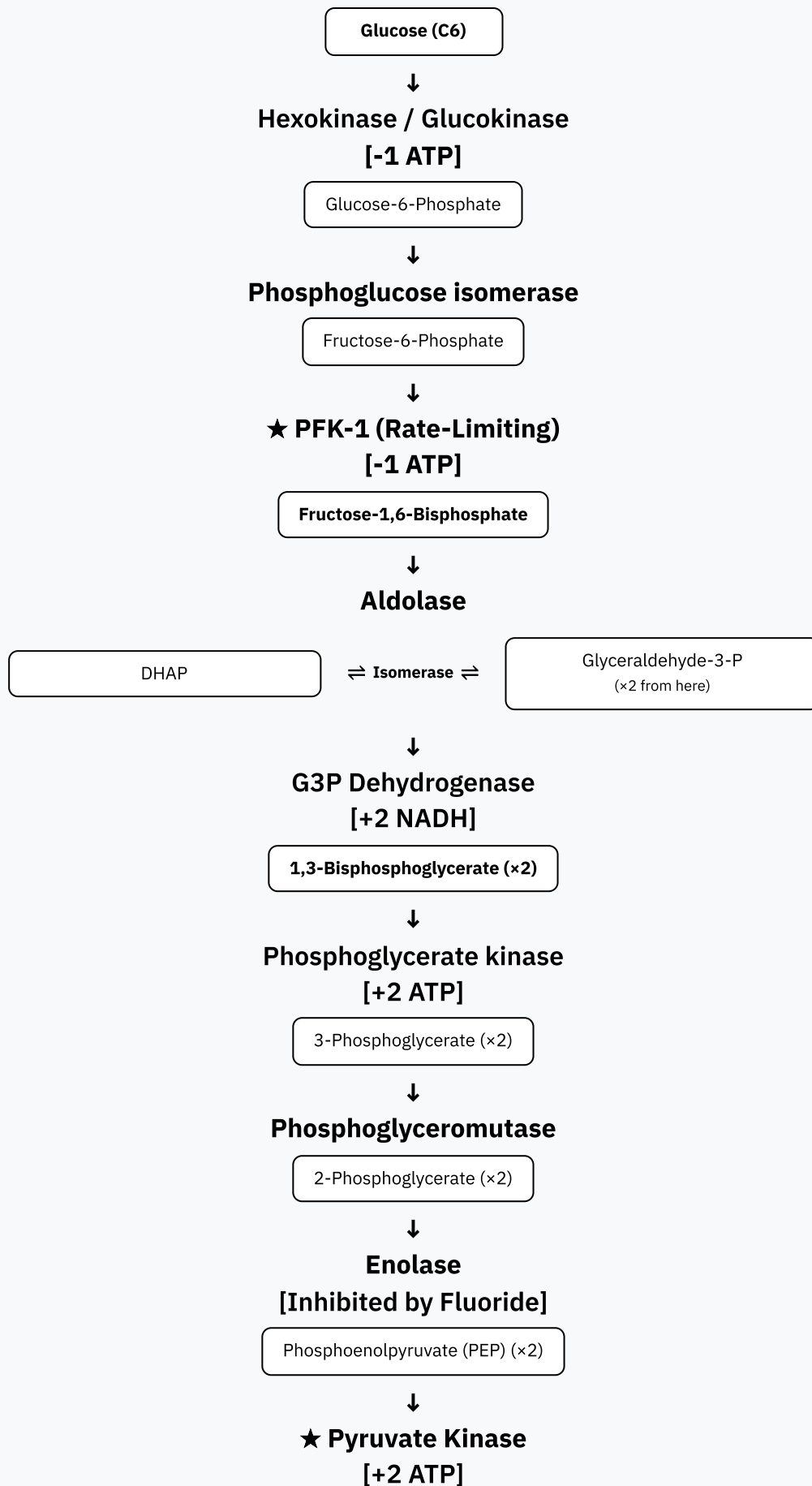


02 Glycolysis ★ EXAMINER FAV

📌 ONE-LINER DEFINITION

Glycolysis is the **cytoplasmic, oxygen-independent** breakdown of one molecule of glucose (6C) into **two molecules of pyruvate (3C)**, yielding a net gain of **2 ATP + 2 NADH**.

// GLYCOLYSIS – 10 STEPS FLOWCHART (EXAM-READY)



Pyruvate (×2) ✓

⚡ ATP invested: 2
ATP gained: 4
Net: +2 ATP

NADH produced: 2
(→ 5 ATP in ETC)

Location: Cytoplasm
O₂ required: X

HEXOKINASE VS. GLUCOKINASE: ISOENZYME COMPARISON

The phosphorylation of glucose to glucose-6-phosphate (Step 1) is catalyzed by distinct isoenzymes depending on the tissue, which represents a crucial control point in carbohydrate homeostasis.

PROPERTY	HEXOKINASE (TYPES I, II, III)	GLUCOKINASE (TYPE IV)
TISSUE DISTRIBUTION	Universal (present in almost all extrahepatic tissues, e.g., muscle, brain).	Localized to liver parenchymal cells and pancreatic β-islet cells .
K_M (AFFINITY FOR GLUCOSE)	Very Low K_m (≈ 0.05 mM) ; represents extremely high affinity. Operates at maximum velocity (V _{max}) even under fasting glucose levels.	Very High K_m (≈ 10 mM) ; represents low affinity. Active only when blood glucose levels rise significantly (postprandial state).
V_{MAX} (CAPACITY)	Low V_{max} ; easily saturated by low concentrations of glucose. Cannot clear large spikes of glucose.	High V_{max} ; high capacity. Allows the liver to rapidly clear massive glucose loads from portal blood after a meal.
FEEDBACK INHIBITION	Allosterically inhibited by its product, Glucose-6-Phosphate (G6P) . Prevents cellular ATP wasting.	Not inhibited by G6P. Active during high glucose to shunt G6P into glycogenesis. Controlled instead by glucokinase regulatory protein (GKRP).
HORMONAL REGULATION	Not regulated by hormones. It is a constitutive basal enzyme.	Induced by Insulin . Insulin increases transcription of the glucokinase gene to promote hepatic glucose utilization.
PHYSIOLOGICAL ROLE	Ensures constant, basal glucose utilization by peripheral organs (e.g., brain) regardless of blood sugar spikes.	Acts as a glucose sensor in pancreatic cells and enables the liver to clear glucose and store it as glycogen after meals.



Rate-Limiting Step:

RATE LIMITING STEP

Phosphofructokinase-1 (PFK-1) — catalyses fructose-6-P → fructose-1,6-bisphosphate. Called the "**pacemaker of glycolysis**." Highly regulated by energy charge.

⚡ ALLOSTERIC REGULATION OF PFK-1 BY FRUCTOSE-2,6-BISPHOSPHATE

PFK-1 is allosterically inhibited by ATP (high-energy signal) and Citrate (TCA intermediate signaling backup fuel). However, **Fructose-2,6-bisphosphate (F-2,6-BP)** acts as the **most potent allosteric activator** of PFK-1. F-2,6-BP binds to PFK-1, dramatically increasing its affinity for Fructose-6-Phosphate and **completely overriding the inhibitory effects of ATP**. This converts PFK-1's kinetic curve from sigmoidal to hyperbolic. F-2,6-BP levels are regulated by the bifunctional enzyme **PFK-2/FBPase-2**, which serves as the primary hormonal switch between glycolysis and gluconeogenesis.



Anaerobic Glycolysis (in RBCs, muscle during sprint): Pyruvate → Lactate (by LDH). This regenerates NAD^+ so glycolysis can continue without oxygen. **RBCs have no mitochondria** — 100% dependent on anaerobic glycolysis for ATP.

// THE RAPOPORT-LUEBERING SHUNT IN RBCS

In mature erythrocytes, a unique metabolic pathway branch bypasses the standard substrate-level phosphorylation step of glycolysis to generate **2,3-Bisphosphoglycerate (2,3-BPG)**. This pathway is crucial for oxygen delivery. It is catalysed by **Bisphosphoglycerate mutase**, which converts 1,3-BPG to 2,3-BPG. 2,3-BPG is then hydrolyzed back to 3-phosphoglycerate by **2,3-BPG phosphatase**, entering the main glycolytic pathway. Because it bypasses the phosphoglycerate kinase reaction, **no ATP is produced** during this shunt, costing the RBC 1 ATP per glucose molecule.

Physiological Role of 2,3-BPG:

- Binds specifically to the central cavity of the hemoglobin tetramer, forming salt bridges that **stabilize the T-state (deoxy-form)** of Hb.
- Decreases hemoglobin's affinity for oxygen, promoting the unloading of oxygen in peripheral tissues. This shifts the **oxygen-hemoglobin dissociation curve to the right**.
- In stored bank blood, 2,3-BPG degrades over time, causing hemoglobin to hold onto oxygen too tightly and reducing tissue oxygenation after transfusion (stored blood paradox).



Clinical Lab Correlation: Fluoride Vials

Blood for glucose estimation is collected in grey-topped fluoride vials containing **Sodium Fluoride (NaF)**. Fluoride acts as an irreversible inhibitor of **Enolase** (Step 9: 2-PG → PEP) by complexing with magnesium (Mg^{2+}) and phosphate, thereby trapping the magnesium cofactor. This completely halts RBC glycolysis, preserving the blood glucose concentration at the time of collection for accurate clinical testing.



High-Yield Toxicological Inhibitors of Glycolysis

1. Iodoacetate & Iodoacetamide: Irreversibly alkylates the essential thiol (-SH) group of the active-site cysteine residue in **Glyceraldehyde-3-Phosphate Dehydrogenase (G3PDH)**, inactivating the enzyme.

2. Arsenite (Trivalent Arsenic): Acts as a structural analog of inorganic phosphate (P_i). It competes with P_i at **G3PDH** and forms an unstable arseno-intermediate (1-arseno-3-phosphoglycerate) which spontaneously hydrolyzes. This uncouples glycolysis, allowing the pathway to proceed without generating ATP at this substrate-level phosphorylation step.

IRREVERSIBLE STEPS OF GLYCOLYSIS (BYPASS POINTS FOR GLUCONEOGENESIS)

Step 1 Glucose → G6P — by **Hexokinase/Glucokinase** (bypassed in GNG by Glucose-6-phosphatase)

Step 3 F6P → F-1,6-BP — by **PFK-1** (bypassed in GNG by Fructose-1,6-bisphosphatase)

Step 10 PEP → Pyruvate — by **Pyruvate Kinase** (bypassed in GNG by Pyruvate carboxylase + PEPCK)

03 Pyruvate Dehydrogenase Complex

THE BRIDGE REACTION: PYRUVATE TO ACETYL-COA

The Pyruvate Dehydrogenase (PDH) complex catalyzes the **irreversible oxidative decarboxylation** of pyruvate in the mitochondrial matrix, forming Acetyl-CoA. This bridge reaction connects cytoplasmic glycolysis directly to the mitochondrial TCA cycle and lipogenesis pathways.

// PDH COMPLEX – STOICHIOMETRY AND SUBUNIT ARCHITECTURE

PDH is not a single enzyme but a massive **multienzyme complex** consisting of multiple copies of three distinct catalytic subunits and requiring **five coenzymes** (derived from B-complex vitamins). This cooperative structural organization prevents the loss of volatile intermediates, maximizing catalytic rate.

THE 3 CATALYTIC SUBUNITS:

E1 – Pyruvate Decarboxylase (Stoichiometry: Tetramer): Catalyzes the decarboxylation of pyruvate. Requires **Thiamine Pyrophosphate (TPP)** (Vitamin B1) as a tightly bound prosthetic group.

E2 – Dihydrolipoyl Transacetylase (Stoichiometry: 24 or 60 subunits, core): Catalyzes the transfer of the acetyl group to Coenzyme A. Requires **Lipoic Acid** (covalently attached as lipoamide) and **Coenzyme A** (pantothenic acid derivative, B5).

E3 – Dihydrolipoyl Dehydrogenase (Stoichiometry: Dimer): Regenerates the oxidized form of lipoamide. Requires **Flavin Adenine Dinucleotide (FAD)** (riboflavin derivative, B2) and **Nicotinamide Adenine Dinucleotide (NAD⁺)** (niacin derivative, B3).

// STEP-BY-STEP CATALYTIC MECHANISM OF PDH

Step 1: Decarboxylation of Pyruvate (E1)

Pyruvate (3C) is decarboxylated by E1, releasing **CO₂**. The remaining 2C hydroxyethyl moiety is covalently attached to TPP, forming *hydroxyethyl-TPP*.

Step 2: Acetyl Group Transfer to Lipoamide (E1 to E2)

The E1 subunit transfers the 2C hydroxyethyl group to the disulfide (oxidized) form of lipoamide on E2. The hydroxyethyl group is oxidized to an acetyl group, reducing the disulfide bond of lipoamide to form *acetyldihydrolipoamide*.

Step 3: Synthesis of Acetyl-CoA (E2)

The E2 subunit transfers the acetyl group from acetyldihydrolipoamide to Coenzyme A (CoA-SH), releasing **Acetyl-CoA** (2C) and leaving the lipoamide in its fully reduced (dihydrolipoamide) state.

Step 4: Oxidation of Dihydrolipoamide (E3)

To allow further catalysis, E3 oxidizes the dihydrolipoamide back to its active disulfide (lipoamide) state. E3's disulfide-bound FAD accepts the two hydrogen atoms, reducing it to **FADH₂**.

Step 5: Electron Transfer to NAD⁺ (E3)

Finally, the FADH₂ on E3 transfers its electrons to **NAD⁺**, generating **NADH + H⁺**, which enters the Electron Transport Chain to produce 2.5 ATP.

✿ SOPHISTICATED PHOSPHORYLATION CASCADE OF PDH COMPLEX

PDH is tightly regulated by allosteric feedback and covalent modification to prevent waste of carbohydrate reserves.

PDH Kinase (Inactivates):

Phosphorylates three specific Serine residues on the E1 subunit, **rendering the complex inactive**. PDH Kinase is **allosterically activated** by ATP, NADH, and Acetyl-CoA (high energy, product accumulation), and **inhibited** by ADP, NAD⁺, CoA-SH, and Pyruvate (low energy, substrate accumulation).

PDH Phosphatase (Activates):

Dephosphorylates the Serine residues, **restoring the complex to its active state**. PDH Phosphatase is **activated by Calcium (Ca²⁺)** (coupling muscle contraction directly to energy generation), Magnesium (Mg²⁺), and **Insulin** (promoting postprandial glucose clearance).

Pathology: Thiamine Deficiency & Arsenic Poisoning

1. Thiamine (Vitamin B1) Deficiency: Leads to lack of TPP, arresting E1 activity. Pyruvate accumulates in the cytoplasm and is converted to lactate by LDH, resulting in **Lactic Acidosis**. Affects tissues highly dependent on aerobic carbohydrate metabolism (brain and heart). Clinical phenotypes include **Wernicke-Korsakoff Syndrome** (nystagmus, ataxia, confabulation, common in alcoholics due to impaired thiamine absorption) and **Beriberi** (Wet: high-output heart failure; Dry: peripheral neuropathy).

2. Arsenic Poisoning (Arsenite): Trivalent arsenite has a high affinity for vicinal sulfhydryl (-SH) groups. It selectively binds and chelates the **lipoic acid** cofactor of the E2 subunit. This permanently arrests both the PDH complex and the α -ketoglutarate dehydrogenase complex (TCA cycle), starving the brain of ATP and presenting with neurological deficits, garlic breath, and hyperkeratosis.

04 TCA / Krebs Cycle ★ EXAMINER FAV

ONE-LINER DEFINITION

The TCA cycle is the **final common pathway** for the oxidation of carbohydrates, lipids, and proteins. It is an **amphibolic** (both anabolic and catabolic), **mitochondrial matrix** cyclic pathway where 1 Acetyl-CoA (2C) condenses with Oxaloacetate (4C) to generate **3 NADH + 1 FADH₂ + 1 GTP + 2 CO₂** per turn.

// TCA CYCLE – COMPLETE FLOWCHART

Oxaloacetate (4C)
+ Acetyl-CoA (2C)

↓ **Citrate Synthase** RLS

Citrate (6C)

↓ **Aconitase**

Isocitrate (6C)

↓ **Isocitrate Dehydrogenase** → CO₂ + NADH

α-Ketoglutarate (5C)

↓ **α-KG Dehydrogenase** → CO₂ + NADH

Succinyl-CoA (4C)

↓ **Succinyl-CoA Synthetase** → GTP ⚡

Succinate (4C)

↓ **Succinate Dehydrogenase** → FADH₂

Fumarate (4C)

↓ **Fumarase** → H₂O added

Malate (4C)

↓ **Malate Dehydrogenase** → NADH

Oxaloacetate (4C) ♻️ regenerated

Per turn:
3 NADH · 1 FADH₂ · 1 GTP · 2 CO₂

ATP yield per turn:
NADH(×3)→7.5 + FADH₂→1.5 + GTP→1
= **10 ATP**

Location:
Mitochondrial matrix

IRREVERSIBLE (REGULATED) STEPS OF TCA

1 Citrate Synthase — Oxaloacetate + Acetyl-CoA → Citrate RLS — Inhibited by: ATP, NADH, Succinyl-CoA, Citrate

3 Isocitrate Dehydrogenase — Activated by: ADP, Ca²⁺. Inhibited by: ATP, NADH

4 α-Ketoglutarate Dehydrogenase — Similar regulation to PDH complex. Needs TPP, Lipoic acid, CoA, FAD, NAD⁺

🔗 AMPHIBOLIC ROLE OF THE TCA CYCLE

The TCA cycle is **amphibolic** because it functions in both catabolism (oxidation of Acetyl-CoA to release ATP energy) and anabolism (intermediates are continuously drawn off as precursors for diverse biosynthetic pathways). This is a high-yield topic frequently tested in MBBS board exams.

TCA INTERMEDIATE	ANABOLIC BIOSYNTHETIC DESTINATION / PATHWAY
CITRATE	Shunted to the cytoplasm to yield Acetyl-CoA for Fatty Acid and Cholesterol/Steroid synthesis.
A-KETOGLUTARATE	Undergoes transamination to yield Glutamate , which is the precursor for other amino acids (proline, arginine) and GABA (neurotransmitter).
SUCCINYL-COA	Condenses with glycine to form δ -aminolevulinic acid (ALA), the first committed step of Heme synthesis .
MALATE	Transported to the cytoplasm to serve as a substrate for gluconeogenesis via oxaloacetate.
OXALOACETATE (OAA)	Undergoes transamination to yield Aspartate , which is used in Purine and Pyrimidine nucleotide synthesis, or enters Gluconeogenesis .



Anaplerosis (Replenishing Reactions): Because cycle intermediates are continuously drawn off for biosyntheses, they must be constantly replenished to keep the cycle running. The primary anaplerotic enzyme is **Pyruvate Carboxylase** (mitochondrial, requires **Biotin / Vitamin B7** and ATP), which directly converts pyruvate to Oxaloacetate. It is allosterically activated by **Acetyl-CoA**—signaling that OAA is depleted and must be replenished to condense with incoming Acetyl-CoA.

💀 Molecular Mechanisms of TCA Cycle Inhibitors

- 1. Fluoroacetate (Suicide Substrate of Aconitase):** Fluoroacetate (commonly used as rodenticide) is not toxic by itself. It condenses with Oxaloacetate to form **Fluorocitrate** via Citrate Synthase. Fluorocitrate then acts as a highly toxic *suicide substrate* that binds covalently to the active site of **Aconitase**, irreversibly blocking the conversion of citrate to isocitrate. This results in massive cellular citrate accumulation and metabolic collapse.
- 2. Malonate (Competitive Inhibitor of Succinate Dehydrogenase):** Malonate is a structural analog of succinate (it is a dicarboxylic acid with 3 carbons, while succinate has 4 carbons). It binds reversibly to the active site of **Succinate Dehydrogenase**, competing directly with succinate. This classic competitive inhibition can be overcome by adding excess succinate.
- 3. Arsenite (Trivalent Arsenic):** Inhibits **α -Ketoglutarate Dehydrogenase** by binding covalently to the sulfhydryl groups of lipoic acid, arresting the cycle at the E2 subunit.

05 HMP Shunt (Pentose Phosphate Pathway)

★ PATHWAYS & PURPOSE OF THE SHUNT

The Hexose Monophosphate (HMP) Shunt is an alternative, cytoplasmic pathway for glucose oxidation that does **not generate or consume ATP**. Instead, it has two major biosynthetic goals: generating **NADPH** for reductive biosyntheses and antioxidant protection, and **Ribose-5-phosphate** for nucleotide synthesis.

// HMP SHUNT – COMPLETE PATHWAY (OXIDATIVE & NON-OXIDATIVE PHASES)

1. The Oxidative Phase (Irreversible):

Glucose-6-Phosphate is oxidized to Ribulose-5-Phosphate. This phase generates exactly **2 molecules of NADPH** and releases 1 molecule of CO_2 per glucose molecule. The rate-limiting enzyme is **Glucose-6-Phosphate Dehydrogenase (G6PD)**.

2. The Non-Oxidative Phase (Reversible Carbon Shuffling):

Ribulose-5-phosphate is interconverted into Ribose-5-phosphate (for DNA/RNA) or recycled back into glycolytic intermediates (Fructose-6-Phosphate and Glyceraldehyde-3-Phosphate) through two key enzymes:

- **Transketolase:** Transfers a 2-carbon ketol group from a ketose to an aldose. **Requires Thiamine Pyrophosphate (TPP / Vitamin B1)** as an essential cofactor. Measuring red blood cell transketolase activity is the gold-standard diagnostic test for Thiamine deficiency.
- **Transaldolase:** Transfers a 3-carbon dihydroxyacetone group. Does not require cofactors.

🚀 HIGH-YIELD FUNCTIONS OF NADPH (EXAM FAVOURITE)

NADPH and NADH are not metabolic equivalents. While NADH is oxidized in the ETC to generate ATP, NADPH is used as a reducing agent in anabolic pathways and cellular defense.

CATEGORY	PHYSIOLOGICAL FUNCTION & BIOCHEMICAL PATHWAYS
1. ANTIOXIDANT DEFENSE	Keeps Glutathione in its reduced (GSH) state via <i>Glutathione Reductase</i> . Reduced glutathione neutralizes highly reactive hydrogen peroxide (H_2O_2) into water via <i>Glutathione Peroxidase</i> , protecting cell membranes from lipid peroxidation.
2. REDUCTIVE BIOSYNTHESIS	Serves as the essential electron donor for Fatty Acid synthesis (in liver, mammary glands, adipose tissue), Cholesterol and Bile Acid synthesis (in liver), and Steroid Hormone synthesis (in adrenal cortex, testes, ovaries).
3. PHAGOCYTTIC BACTERICIDAL ACTIVITY	Used by NADPH Oxidase in neutrophils and macrophages to generate superoxide radicals ($O_2^{\bullet -}$), initiating the respiratory burst that kills engulfed bacteria. Defect in NADPH oxidase leads to <i>Chronic Granulomatous Disease</i> .
4. NITRIC OXIDE (NO) SYNTHESIS	Required as a co-substrate by <i>Nitric Oxide Synthase</i> (NOS) to produce NO from arginine, promoting vasodilation and neurotransmission.

🧬 Molecular Pathology: G6PD Deficiency (Classic Case Study)

G6PD deficiency is the **most common clinically significant enzyme deficiency** in the world, affecting over 400 million people (X-linked recessive, offering evolutionary protection against *Plasmodium falciparum* malaria).

The Erythrocyte Vulnerability:

Mature erythrocytes **lack nuclei, ribosomes, and mitochondria**. They cannot synthesize new proteins. Therefore, any G6PD enzyme degraded over the RBC's 120-day lifespan cannot be replaced. The RBC is 100% dependent on G6PD as its *only* source of NADPH to maintain reduced glutathione.

The Cascade of Hemolytic Crisis:

Under oxidative stress (triggered by **Primaquine** antimalarials, **Dapsone**, **Fava beans** ingestion, or severe **infection**), reactive oxygen species (ROS) accumulate inside the RBC due to lack of reduced glutathione (GSH).

ROS Accumulates → Lipid Peroxidation of Membrane → Methemoglobin Aggregates (Heinz Bodies)
→ Splenic Macrophages Pluck Out Aggregates ("Bite Cells") → Intravascular & Extravascular Hemolysis

Diagnostic Lab Findings:

1. Heinz bodies on peripheral blood smear (crystal violet stain).
2. "Bite cells" on peripheral smear.
3. Hemoglobinuria (dark tea-colored urine) and increased indirect bilirubin (jaundice).
4. Quantitative assay showing decreased G6PD enzyme activity (Note: must be checked *weeks after* the hemolytic episode, as active hemolysis clears older G6PD-deficient RBCs first, leading to a false normal).

06 Glycogen Metabolism ★ EXAMINER FAV

★ CELLULAR GLYCOGEN DYNAMICS

Glycogen is a highly branched polymer of glucose stored in liver (maintains blood glucose) and skeletal muscle (reservoir for contraction). Synthesis (glycogenesis) and breakdown (glycogenolysis) are tightly regulated reciprocal pathways that never occur simultaneously in the same cell.

// THE GLYCOGENIN PRIMER MECHANISM IN GLYCOGENESIS

Glycogen Synthase, the rate-limiting enzyme of glycogenesis, can only add glucose residues (forming α -1,4 links) to an **existing chain of at least 8 glucose units**. It cannot initiate synthesis de novo. Initiation requires **Glycogenin**, a specialized core protein that acts as an enzyme:

1. **Auto-glucosylation:** Glycogenin catalyzes the transfer of a glucose residue from UDP-Glucose to the hydroxyl group of its own **Tyrosine-194 residue**.
2. **Primer Extension:** It then sequentially adds up to 7 more glucose residues from UDP-Glucose to form a short, linear glucan primer. Once this primer of 8 units is complete, **Glycogen Synthase** takes over to rapidly elongate the chain.

// GLYCOGENESIS & GLYCOGENOLYSIS PATHWAYS

GLYCOGENESIS (SYNTHESIS):

Glucose $\rightarrow$ Glucose-6-P $\rightarrow$ Glucose-1-P.
Condenses with UTP via *UDP-Glucose Pyrophosphorylase* to form UDP-Glucose. **Glycogen Synthase** (RLS) transfers glucose from UDP-Glucose to the primer (α -1,4 linkages). Once the chain reaches 11-12 residues, **Branching Enzyme** (amylo- α -1,4 $\rightarrow$ α -1,6 transglucosidase) transfers a block of 7 residues to a neighboring chain, creating a branch point via an **α -1,6-glucosidic bond**.

GLYCOGENOLYSIS (BREAKDOWN):

Glycogen Phosphorylase (RLS) cleaves α -1,4 linkages phosphorolytically using P_i , yielding Glucose-1-Phosphate (G1P), until 4 residues remain from a branch point (called *Limit Dextrin*). **Debranching Enzyme** (bifunctional) then resolves it: its *transferase* domain moves 3 glucosyl units to an adjacent chain, and its *α -1,6-glucosidase* domain hydrolyzes the single remaining branch glucose, releasing **free Glucose**.

// RECIPROCAL HORMONAL REGULATION CASCADE (cAMP/PKA PATHWAY)

Glucagon (in liver during fasting) or Epinephrine (in muscle during flight-or-fight) binds to GPCRs, activating Adenyl Cyclase → **cAMP levels rise**, activating **Protein Kinase A (PKA)**. PKA phosphorylates key enzymes to orchestrate a coordinated metabolic response:

1. Glycogen Synthase is Phosphorylated & Inactivated:

PKA directly phosphorylates Glycogen Synthase, converting it from its active dephosphorylated form (*Synthase a*) to its inactive phosphorylated form (*Synthase b*). **Glycogenesis is arrested.**

2. Phosphorylase Kinase is Phosphorylated & Activated:

PKA phosphorylates Phosphorylase Kinase, activating it. Active Phosphorylase Kinase then phosphorylates Glycogen Phosphorylase, converting it from its inactive dephosphorylated form (*Phosphorylase b*) to its highly active phosphorylated form (*Phosphorylase a*). **Glycogenolysis is activated.**

3. Protein Phosphatase 1 (PP1) is Inhibited:

PKA phosphorylates *Inhibitor-1*, which binds to and inactivates PP1, preventing it from dephosphorylating (and reversing) these active enzymes.

The Insulin Reversal: Insulin binds to tyrosine kinase receptors → activates phosphodiesterase (degrades cAMP) and activates **Protein Phosphatase 1 (PP1)** → PP1 dephosphorylates both enzymes → activates Glycogen Synthase and inactivates Glycogen Phosphorylase → promotes Glycogenesis.

COMPREHENSIVE PATHOLOGY OF GLYCOGEN STORAGE DISEASES (GSDS)

Genetic deficiencies of glycogen metabolizing enzymes cause GSDs, resulting in abnormal glycogen accumulation and metabolic deficits. This is a classic Group B/C MBBS question.

GSD TYPE & NAME	DEFECTIVE ENZYME	ORGANS AFFECTED	CLINICAL PRESENTATION & BIOCHEMICAL RATIONALE
TYPE I: VON GIERKE'S DISEASE	Glucose-6-phosphatase (G6Pase)	Liver & Kidney	Severe fasting hypoglycemia (cannot release glucose from liver glycogen or gluconeogenesis). Hepatomegaly , lactic acidosis (excess G6P enters glycolysis to yield lactate), and hyperuricaemia/gout (accumulated G6P shunted into HMP pathway → excess Ribose-5-P → purines → uric acid).
TYPE II: POMPE'S DISEASE	Lysosomal α -1,4-glucosidase (Acid Maltase)	All organs (especially Heart, Muscle)	Lysosomal storage disease . Glycogen accumulates inside lysosomal vacuoles. Leads to massive cardiomegaly , generalized hypotonia ("floppy baby"), and early death due to heart failure. Normal blood glucose.
TYPE III: CORI'S DISEASE	Debranching enzyme (α -1,6-glucosidase)	Liver & Muscle	Milder fasting hypoglycemia. Glycogen molecules with abnormally short outer branches (limit dextrins) accumulate in tissues. Hepatomegaly and muscle hypotonia.
TYPE IV: ANDERSEN'S DISEASE	Branching enzyme (amylo-1,4 → 1,6-transglucosidase)	Liver & Spleen	Infantile liver cirrhosis and portal hypertension. Glycogen accumulates as long, linear, unbranched polymers with very low solubility, triggering a foreign body immune reaction in hepatocytes. Fatal by age 2.
TYPE V: MCARDLE'S DISEASE	Muscle Glycogen Phosphorylase	Skeletal Muscle	Exercise-induced muscle cramps , fatigue, and transient myoglobinuria (rhabdomyolysis) leading to red-colored urine. No rise in blood lactate after strenuous exercise (forearm ischemic exercise test is diagnostic) because muscle cannot break down glycogen.
TYPE VI: HERS' DISEASE	Liver Glycogen Phosphorylase	Liver	Mild hepatomegaly and mild fasting hypoglycemia. Excellent prognosis.

07 Gluconeogenesis

★ DEFINITION & BIOSYNTHETIC PURPOSE

Gluconeogenesis (GNG) is the synthesis of **new glucose from non-carbohydrate precursors**. It occurs primarily in the **liver (90%)** and to a lesser extent in the kidney cortex (10%) during fasting and starvation. GNG is not simply the reverse of glycolysis; the three highly exergonic, irreversible steps of glycolysis must be bypassed by unique mitochondrial, cytosolic, and endoplasmic reticulum enzymes.



Gluconeogenic Precursors:

- **Lactate:** Derived from anaerobic glycolysis in RBCs and exercising skeletal muscle.
- **Glucogenic Amino Acids:** Derived from muscle protein breakdown during fasting (primarily **Alanine**, which is converted to pyruvate).
- **Glycerol:** Derived from adipocyte triacylglycerol hydrolysis (phosphorylated to glycerol-3-phosphate and oxidized to DHAP).
- **Propionyl-CoA:** Derived from odd-chain fatty acid oxidation (converted to Succinyl-CoA). **Even-chain fatty acids cannot yield glucose** because the PDH reaction is irreversible and Acetyl-CoA cannot be converted back to pyruvate in net amounts.

// THE 3 HIGHLY COMPARTMENTALIZED BYPASS STEPS

Bypass 1: Pyruvate to Phosphoenolpyruvate (PEP) – Mitochondrial & Cytosolic

1. **Mitochondrial Pyruvate Carboxylase (PC):** Pyruvate is transported into the mitochondria and carboxylated to Oxaloacetate (OAA). Requires **Biotin (Vitamin B7)**, ATP, and CO₂. PC is allosterically activated by **Acetyl-CoA**.
2. **Malate Shuttle:** Because OAA cannot cross the inner mitochondrial membrane, it is reduced to Malate by mitochondrial *Malate Dehydrogenase*, transported to the cytoplasm, and re-oxidized back to OAA.
3. **Cytosolic PEPCK:** Cytosolic *Phosphoenolpyruvate Carboxykinase* decarboxylates and phosphorylates OAA to PEP, requiring **GTP** as a phosphate donor.

Bypass 2: Fructose-1,6-Bisphosphate to Fructose-6-Phosphate – Cytosolic

Catalyzed by **Fructose-1,6-Bisphosphatase-1 (FBPase-1)** via simple hydrolytic cleavage. This is the key regulatory step of gluconeogenesis. FBPase-1 is **allosterically activated by Citrate** and **strongly inhibited by AMP** and **Fructose-2,6-bisphosphate**.

Bypass 3: Glucose-6-Phosphate to free Glucose – Endoplasmic Reticulum

Catalyzed by **Glucose-6-Phosphatase (G6Pase)**. G6Pase is an integral membrane protein bound to the **lumen of the endoplasmic reticulum**. G6P is transported into the ER lumen, cleaved, and the resulting free glucose is exported via GLUT2. G6Pase is **absent in skeletal muscle**, explaining why muscle glycogen cannot maintain blood glucose during fasting.

- ⚡ **Thermodynamics & Energetic Cost:** Gluconeogenesis is a highly endergonic pathway. The synthesis of **1 molecule of glucose from 2 molecules of pyruvate costs exactly 6 high-energy phosphate bonds: 4 ATP** (2 at Pyruvate Carboxylase, 2 at Phosphoglycerate Kinase) + **2 GTP** (at PEPCK). This energy is supplied by hepatic β -oxidation of fatty acids during fasting.

// INTER-ORGAN METABOLIC EXCHANGE CYCLES

The Cori Cycle (Lactate Exchange):

Anaerobic glycolysis in RBCs/exercising muscle generates **Lactate**. Lactate is released into blood, taken up by the liver, and oxidized to pyruvate. Liver GNG converts pyruvate back to glucose, which is released into the blood to be utilized by muscle again. Net: shifts the energetic cost of lactate disposal (6 ATP) from oxygen-depleted muscle to the liver.

The Glucose-Alanine Cycle (Nitrogen Exchange):

Active muscle breaks down proteins to amino acids. The amino groups are transaminated to pyruvate, forming **Alanine**. Alanine travels to the liver, where it is transaminated back to pyruvate (which enters GNG) and the amino group enters the **Urea Cycle** for safe excretion. Net: safely disposes of muscle nitrogen while replenishing blood glucose.

✿ PFK-2/FBPASE-2 BIFUNCTIONAL ENZYME: THE ALLOSTERIC COORDINATOR

Glycolysis and GNG are reciprocally regulated by **Fructose-2,6-bisphosphate (F-2,6-BP)**. The cellular level of F-2,6-BP is controlled by a single homodimeric protein with **two opposing catalytic domains**: PFK-2 and FBPase-2.

FED STATE (Insulin Dominated):

High insulin activates *Protein Phosphatase*, dephosphorylating the bifunctional enzyme. This **activates the PFK-2 domain** and inactivates the FBPase-2 domain. F-2,6-BP levels **rise**, strongly activating PFK-1 and inhibiting FBPase-1. **Glycolysis is stimulated; GNG is shut down.**

FASTING STATE (Glucagon Dominated):

High glucagon increases cAMP, activating *Protein Kinase A (PKA)*. PKA phosphorylates the bifunctional enzyme. This **inactivates the PFK-2 domain** and **activates the FBPase-2 domain**. F-2,6-BP is degraded. The drop in F-2,6-BP relieves the inhibition on FBPase-1 and removes PFK-1 activation. **GNG is stimulated; Glycolysis is arrested.**

08 Blood Glucose Homeostasis

// FED VS FASTING STATE METABOLIC PROFILES

FED STATE (↑ Insulin)

- Glucose uptake ↑ (GLUT4 in muscle/fat)
- Glycolysis ↑ (aerobic energy)
- Glycogenesis ↑ (liver/muscle storage)
- Lipogenesis ↑ (excess glucose → fatty acids)
- Gluconeogenesis ↓
- Protein synthesis ↑

FASTING STATE (↑ Glucagon)

- Glycogenolysis ↑ (liver glycogen maintains blood glucose first 12-18h)
- Gluconeogenesis ↑ (takes over fully after glycogen depletion)
- Lipolysis ↑ (FFA release as alternative fuel)
- Ketogenesis ↑ (supplies brain energy during starvation)
- Glycolysis ↓

09 Fructose & Galactose Metabolism

MOLECULAR PATHOLOGY OF FRUCTOSE METABOLISM

Fructose is metabolized primarily in the liver. It bypasses the rate-limiting PFK-1 step of glycolysis, explaining why high-fructose diets can rapidly drive hepatic lipogenesis.

Normal Pathway: Fructose is phosphorylated to Fructose-1-Phosphate (F1P) by **Fructokinase**. F1P is then cleaved by **Aldolase B** into dihydroxyacetone phosphate (DHAP) and glyceraldehyde, which enter glycolysis.

Hereditary Fructose Intolerance (HFI) — Aldolase B Deficiency:

Deficiency of Aldolase B leads to the accumulation of **Fructose-1-Phosphate (F1P)** in hepatocytes. This traps massive amounts of cellular **inorganic phosphate (P_i)**, depleting ATP pools.

Aldolase B Defect → F1P Accumulates → Inorganic Phosphate (P_i) Trapped → ATP Depletion
→ Inhibition of Glycogenolysis (needs P_i) and Gluconeogenesis (needs ATP) → Severe Hypoglycemia

Clinical Presentation: Severe fasting hypoglycemia, vomiting, and hepatomegaly appearing upon the introduction of sucrose, fructose, or fruit juices to an infant's diet. Chronic exposure leads to hepatic and renal failure.

Essential Fructosuria — Fructokinase Deficiency:

A completely benign, asymptomatic autosomal recessive condition. Because fructokinase is missing, fructose cannot be trapped in hepatocytes and is simply excreted in the urine. Spotted incidentally as a reducing sugar in urine.

MOLECULAR PATHOLOGY OF GALACTOSE METABOLISM

Galactose (derived from dietary lactose) is metabolized via the Leloir pathway in the liver. Disruption here leads to toxic metabolic accumulations.

Normal Leloir Pathway: Galactose is phosphorylated to Galactose-1-Phosphate (Gal-1-P) by **Galactokinase**. Gal-1-P is converted to UDP-galactose by **Galactose-1-Phosphate Uridyl Transferase (GALT)**, which is then epimerized to UDP-glucose to enter glycogenesis or glycolysis.

Classical Galactosemia — GALT Deficiency:

Deficiency of GALT leads to accumulation of **Galactose-1-Phosphate** and **Galactose** in tissues. This traps phosphate (similar to HFI) causing liver cirrhosis and renal failure. The accumulated galactose is shunted to **Galactitol** by *Aldose Reductase*. Galactitol accumulates in the avascular lens, causing osmotic swelling and **cataracts**.

Clinical Presentation: Jaundice, hepatomegaly, infantile cataracts, feeding difficulties, and intellectual disability appearing within days of breastfeeding (lactose ingestion). Mandatory newborn screening target.

Galactokinase Deficiency:

A milder disorder. Galactose accumulates but Gal-1-P does not, so there is no toxic phosphate trapping. Presenting symptoms are primarily limited to **infantile cataracts** (due to galactitol accumulation in the lens) and galactosemia/galactosuria.

10 Memory Mnemonics

MNEMONIC 1 — TCA INTERMEDIATES IN ORDER

"Citrate Is Krebs' Starting Substrate For Making Oxaloacetate"

Citrate → Isocitrate → α-Ketoglutarate → Succinyl-CoA → Succinate → Fumarate → Malate → Oxaloacetate

MNEMONIC 2 — IRREVERSIBLE GLYCOLYSIS STEPS

"Hungry People Feel Pale" → HK, PFK-1, PK

Hexokinase (step 1) · Phosphofructokinase-1 (step 3) · Pyruvate Kinase (step 10) — these 3 are irreversible and are bypassed in gluconeogenesis.

MNEMONIC 3 — PDH COMPLEX COENZYMES

"The Lovely Cat Finds Nice Accommodation"

TPP (B1) · Lipoic acid · CoA (B5 pantothenic acid) · FAD (B2) · NAD⁺ (B3) · Note: All B-vitamins!

🧠 MNEMONIC 4 – GLYCOGEN STORAGE DISEASES

"Very Poor Carbohydrate Metabolism" → Types I, II, III, V

Type I = **V**on Gierke (G6Pase) · Type II = **P**ompe (Acid maltase) · Type III = **C**ori (Debranching) · Type V = **M**cArdle (Muscle phosphorylase)

🧠 MNEMONIC 5 – GNG PRECURSORS

"GALA Party" → Glycerol, Amino acids, Lactate, Alanine / Propionate

All glucogenic amino acids → glucose. Fatty acids (even-chain) CANNOT → glucose (no net conversion from Acetyl-CoA).
Odd-chain FA → Propionyl-CoA → Succinyl-CoA → CAN enter GNG.

11 Clinical Correlations

🔗 High-Yield Clinical Links

- **Diabetes Mellitus:** Insulin deficiency → ↓ GLUT4 → ↓ glucose uptake → Hyperglycaemia. ↑ Glycogenolysis, ↑ GNG, ↑ Lipolysis, ↑ Ketogenesis → Diabetic Ketoacidosis (DKA). HbA1c reflects average glucose over 3 months (glycated Hb).
- **Lactic Acidosis:** Anaerobic glycolysis excess (shock, sepsis) or PDH deficiency → Lactate accumulates → ↓ pH. High Anion Gap Metabolic Acidosis.
- **Von Gierke's + Hyperuricaemia:** ↑ G6P → ↑ HMP shunt → ↑ Ribose-5-P → ↑ Purine synthesis → ↑ Uric acid. Classic PYQ trap — explain the link!
- **Galactosaemia → Cataract:** Galactose → Galactitol (by aldose reductase) → Accumulates in lens (avascular) → Osmotic damage → Cataract. Treat: eliminate galactose/lactose from diet.
- **Uncontrolled DM → Cataract:** Glucose → Sorbitol (aldose reductase) → Accumulates in lens → Same mechanism as galactosaemia.
- **Primaquine + G6PD deficiency:** Oxidative drug → ↑ ROS → ↓ NADPH (G6PD absent) → ↓ GSH → Heinz bodies → Haemolysis. Tested in clinical MCQs.

VIVA RAPID FIRE – CARBOHYDRATE METABOLISM

Q1 Why can't RBCs use the TCA cycle?

RBCs have no mitochondria — TCA cycle and oxidative phosphorylation are mitochondria-dependent. RBCs rely entirely on anaerobic glycolysis for ATP. They also use HMP shunt to generate NADPH for maintaining reduced glutathione (protection against oxidative damage).

Q2 Why does Von Gierke's disease cause hyperuricaemia?

G6Pase deficiency → G6P accumulates → excess G6P shunted into HMP pathway → excess Ribose-5-phosphate → increased de novo purine synthesis → excess purines → increased uric acid production → hyperuricaemia and gout.

Q3 Why is gluconeogenesis NOT the simple reverse of glycolysis?

Three irreversible steps of glycolysis cannot be reversed. These are catalysed by Hexokinase, PFK-1, and Pyruvate Kinase. GNG uses unique bypass enzymes: Glucose-6-phosphatase, Fructose-1,6-bisphosphatase, and the Pyruvate Carboxylase + PEPCK system respectively.

Q4 What is the role of 2,3-BPG? Why is it important in stored blood?

2,3-BPG binds to the β -chains of deoxy-Hb, stabilising the T (tense/deoxy) form, reducing O_2 affinity → promotes O_2 delivery to tissues. In stored blood (bank blood), 2,3-BPG levels fall → Hb has increased O_2 affinity → poor O_2 delivery to recipient's tissues (stored blood paradox).

Q5 Why is TCA called amphibolic?

Amphibolic = both anabolic and catabolic. Catabolic: oxidises Acetyl-CoA → CO_2 + energy. Anabolic: intermediates are precursors for biosynthesis — α -KG → glutamate → amino acids; Succinyl-CoA → haem synthesis; OAA → gluconeogenesis; Citrate → fatty acid synthesis.

Q6 Why does muscle lack glucose-6-phosphatase but liver has it?

Liver's role is to maintain blood glucose homeostasis — it must release free glucose into circulation during fasting. G6Pase allows dephosphorylation of G6P → free glucose → export. Muscle is a glucose consumer — it needs G6P to stay within the cell for glycolysis. If muscle had G6Pase, glucose would leak out rather than be used for energy.

Q7 Fat can be synthesised from glucose but glucose cannot be synthesised from fat — justify.

Glucose → Acetyl-CoA (irreversible via PDH). Fatty acids → β -oxidation → Acetyl-CoA. Acetyl-CoA cannot be converted back to pyruvate/OAA in net amounts in mammals — the PDH reaction is irreversible. Acetyl-CoA can only enter TCA (catabolic) or be used for lipid synthesis, not for GNG. Exception: glycerol backbone of TAG can yield glucose (via DHAP).

Q8 What is the significance of oral glucose in ORS (Oral Rehydration Solution)?

Glucose in ORS co-transporters sodium via SGLT1 (Na^+ -glucose co-transporter) in the intestinal mucosa — a secondary active transport system. This Na^+ uptake drags water with it by osmosis, rehydrating the patient. Glucose doesn't just provide energy — its primary role in ORS is to drive Na^+ absorption even in secretory diarrhoea where other Na^+ transporters are inhibited.

13 Group-D EQ Template

 HOW TO ANSWER: "PHOSPHOFRUCTOKINASE-I IS THE PACEMAKER OF GLYCOLYSIS"

Use your teacher's flowchart formula — Define → Mechanism Flowchart → Kinetics/Regulation → Clinical consequence.

// EQ ANSWER FLOWCHART

Define "Pacemaker"

Rate-limiting, essentially irreversible, most regulated step

**Reaction**
$$\text{F6P} + \text{ATP} \rightarrow \text{F-1,6-BP} + \text{ADP}$$

(catalysed by PFK-1)

**Activators**AMP, ADP, F-2,6-BP, Pi
(signals: low energy)**Inhibitors**ATP, Citrate, low pH
(signals: high energy)**Clinical relevance**F-2,6-BP (insulin ↑, glucagon ↓)
coordinates glycolysis vs GNG**MBBS Biochemistry – Module 02: Carbohydrate Metabolism**

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